

Session 221: Communication Signal Processing and Coding

TIME 2:30 PM – 4:00 PM IST VENUE SP 1.20 (TVS Lab)



221.1

Hardware-Efficient FPGA Implementation of OTSM Transceiver for High-Mobility 6G Communications

Vishnupriya P V, Sapta Girish Neelam

221.2

Iteratively reweighted algorithms for robust sparse recovery with L_p ($0 < p < 1$)-norm minimization

Shweta Bharti, Samrat Mukhopadhyay, Govind Murmu

221.3

Coded SumComp: Linear Coding over Integer Rings for Improved SumComp Performance

Vaishnavi Anekar, Anjana A M

221.4

A Novel Tunable Window for Signal Processing Applications

Debanjan Bhattacharjee, Sujit Kumar Sahoo

221.5

PYNQ-Based Modular Architecture for Real-Time Stream-Cipher Video Encryption at the Edge

Vanukuru Hari Rohit, D. Sreekirana, Raghavendra Kune, D. Mallikharjuna Rao, Novaline Jacob

221.6

GSM-Based Non-Orthogonal Multiple Access: Design and Evaluation

Surjeet Kumar, Ashish Kant Shukla, Manav R Bhatnagar

221.7

Enhancing Power Efficiency in MIMO DFT-s-OFDM: A Relative Study of $\pi/2$ -BPSK with Tone Reservation

Abhinaba Dey, Sreenath Ramanath

221.8

Molecular Communication with Finite-Length Transmitter in Time Dependent Diffusive Channel

Neha Bhusari, Avinash G. Keskar, Nikhil Sharma, Prabhat Kumar Sharma

221.9

Generalized Partial Full-Pilot Zero-Forcing Combining for Distributed Massive MIMO

Mohd Saif Ali Khan, Karthik RM, Samar Agnihotri

221.10

Signal Space Decoupled Stream Splitting for Low Latency Downlink MIMO WBANs

Gunjit Arora, Satya Kumar Vankayala, Manoj Thomas

221.11

Low Complexity Digital Predistortion for Power Amplifiers Using Incremental PCA

Girish Tripathi, Anindya Saha

221.12

Rank-Deficit Hybrid Adaptive Fulcrum Decoder for Low-Complexity Network Coding

Sovanjyoti Giri, Satya Prakash

221.13

Evolutionary Optimization for Quantized RIS Phase Shift Design in MIMO Systems: Hyperparameter Analysis and SNR Performance

Amit Shrivastava, Manish Kumar, Kunwar Pritiraj Rajput

221.14

Deep Unfolded AMP for Short Block-Length SPARC over AWGN Channels

Sai Dinesh Kancharana, Sandeep VSV, Arun Pachai Kannu

221.15

Detection in Ambient Backscatter Communication System with Multi-Antenna Tag-Reader Pair with Correlated Ambient Source Signal

Anuranjan Jha, Adarsh Patel

Session 222: Vision and Sensor Analytics

TIME 2:30 PM – 4:00 PM IST VENUE SP Building Side Foyer



222.1 Signal-driven Shipwreck Segmentation and Probabilistic Uncertainty Estimation with Sidescan Sonar for Autonomous Underwater Navigation

Lakshmi Annamalai

222.2 Vision-Powered Cleaning Mode Detection in Robot Vacuum Cleaners Using a Novel Smart Home Dataset

Sanket Gharatkar, Suhel Khan, Nagaraj Yendeti

222.3 NuSegFormer: Enhanced Cell Nuclei Segmentation Using a Hybrid UNet--Vision Transformer with Gated Multi-Head Attention

Aditi Damodar, Hitesh Tekchandani, Praveen Kumar Shukla, Anish Kumar Vishwakarma

222.4 On the Optimal Number of Grids for Differentially Private Non-Interactive K-Means Clustering

Gokularam Muthukrishnan, Anshoo Tandon

222.5 Are Wavelets More Reliable? Comparing The Topology of Image Generator Metrics

Manas Patil, Amit Tiwari, Abhijat Bharadwaj, Vikram Gadre

222.6 Domain-Adaptive Self-Supervised Pretraining for Bacterial Flagellar Motor Detection in Cryo-Electron Tomography

Adavi Jayanth, Madhu Oruganti

222.7 Enhancing Road Safety through Real-Time Unsupervised Anomaly Detection in Crash Barriers

Divya Varshney, Avnish Panwar, Indrajit Ghosh

222.8 Multi-Sensor Alignment for Weather Simulations

Samsad Alam, Devyani Lambhate, Aditya Mohan, Vishal Kumar, Vaibhav Katewa

222.9 SIADS: Innovation-Based Anomaly Detection in Streaming Communication Graphs

Tanish Verma

222.10 LS-PolypSeg: A Parameter-Efficient LoRA-Enhanced SAM3 Framework for Polyp Segmentation

Sanjana Jhansi Ganji, Srikanth Panigrahi, Kaushal Sambanna, Srinivas Rao Routhu

222.11 Tube Loss-based Detection of Anomalies in Electricity Consumption Time Series Data

Shruti Mehta, Pritam Anand, Yash Vasavada

222.12 Learning What to Trust: Confidence-Aware Curriculum Distillation for TinyML Vision

Satarupa Das, Soumen Garai, Rajrup Saha, Soumya Chatterjee, Suman Samui

Session 223: Wireless Resource Optimization

TIME 2:30 PM – 4:00 PM IST VENUE ECE 1.07



223.1

Mobile Base Station Optimal Tour in Wide Area IoT Sensor Networks

Sachin Kadam

223.2

Robust Deep Reinforcement Learning for Task Scheduling in MEC-Enabled IoT Networks

Vikky Masih, Arghyadip Roy

223.3

DPAS: Demand-Pressure Adaptive Slicing for Real-Time 6G Network Resource Management

Jaynil Panjwani, Neha Agrawal

223.4

Ergodic Throughput in Coordinated Multi-point Downlink under Distributed Power Constraints

mahesh nagulapalli, Mahendra Pratap Singh, Anil Kumar, Sibi Raj B. Pillai

223.5

Mobility-Aware Offloading with SINR Prediction and Selective Replication in Fog-IoT Networks

Rajasekhar Dasari, Satyajit Mohapatra, Sanjeet Kumar Nayak

223.6

Decoding-Order-Aware Secrecy Rate Maximization in IRS-Assisted Untrusted NOMA Under Imperfect SIC

Ankit Kumar, Kaneez Khatoon, Deepak Mishra, Ravikant Saini

223.7

Reinforcement Learning-Based Decode and Forward UAV Relay Trajectory Optimization

aniket Subbanwar, Ojas Joshi, Amit Agarwal

223.8

Measurement-Based Energy-Efficiency Optimization for On-demand Data Rate Provisioning in 5G RAN

Ramagiddaiah Eediga, Preethi G V, Anand S V R, Chandra R. Murthy, Ryan Husbands

223.9

Efficient Sampling for Pure Exploration Problems

Dhinesh Chandran G, Srinivas Reddy Kota, Srikrishna Bhashyam

223.10

Link Adaptation Using Joint-Thompson Sampling

Vignatha Vinjam, Manjunath Kolavennu, Myna Vajha, Karthik Periyapattana Narayanaprasad

223.11

Adaptive Scheduling for Low-Latency Coordination in Wi-Fi 8 Multi-AP Networks

Hiya Shah

223.12

Robust Base Station Placement in Agricultural IoT via Bayesian Optimization

Gourav Prateek Sharma, Durgesh Singh, James Gross